

Algoritmos - Actividad Guiada 3

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URL: <https://colab.research.google.com/drive/1Lhithsm51QHS66ZvecNXjEmXminUI0c?usp=sharing>

GitHub: <https://github.com/epalominola/AlgoritmoOptimizacion>

Carga de librerías

```
In [ ]: !pip install requests      #Hacer llamadas http a paginas de la red
        !pip install tsplib95     #Modulo para las instancias del problema del TSP
```

```

Requirement already satisfied: requests in /usr/local/lib/python3.13/dist-packages (2.32.4)
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.13/dist-packages (from requests) (3.4.9)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.13/dist-packages (from requests) (3.19)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.13/dist-packages (from requests) (2.5.0)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.13/dist-packages (from requests) (2026.7.22)
Collecting tsplib95
  Downloading tsplib95-0.7.1-py2.py3-none-any.whl.metadata (6.3 kB)
Requirement already satisfied: Click>=6.0 in /usr/local/lib/python3.13/dist-packages (from tsplib95) (8.5.0)
Collecting Deprecated~1.2.9 (from tsplib95)
  Downloading Deprecated-1.2.18-py2.py3-none-any.whl.metadata (5.7 kB)
Collecting networkx~2.1 (from tsplib95)
  Downloading networkx-2.8.8-py3-none-any.whl.metadata (5.1 kB)
Collecting tabulate~0.8.7 (from tsplib95)
  Downloading tabulate-0.8.10-py3-none-any.whl.metadata (25 kB)
Collecting wrapt<2,>=1.10 (from Deprecated~1.2.9->tsplib95)
  Downloading wrapt-1.17.3-cp313-cp313-manylinux1_x86_64.manylinux2_28_x86_64.manylinux2_5_x86_64.whl.metadata (6.4 kB)
  Downloading tsplib95-0.7.1-py2.py3-none-any.whl (25 kB)
  Downloading Deprecated-1.2.18-py2.py3-none-any.whl (10.0 kB)
  Downloading networkx-2.8.8-py3-none-any.whl (2.0 MB)
  

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 2.0/2.0 MB 21.1 MB/s eta 0:00:00
  Downloading tabulate-0.8.10-py3-none-any.whl (29 kB)
  Downloading wrapt-1.17.3-cp313-cp313-manylinux1_x86_64.manylinux2_28_x86_64.manylinux2_5_x86_64.whl (88 kB)
  

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 88.1/88.1 kB 5.2 MB/s eta 0:00:00
Installing collected packages: wrapt, tabulate, networkx, Deprecated, tsplib95
  Attempting uninstall: wrapt
    Found existing installation: wrapt 2.3.0
    Uninstalling wrapt-2.3.0:
      Successfully uninstalled wrapt-2.3.0
  Attempting uninstall: tabulate
    Found existing installation: tabulate 0.9.0
    Uninstalling tabulate-0.9.0:
      Successfully uninstalled tabulate-0.9.0
  Attempting uninstall: networkx
    Found existing installation: networkx 3.6.1
    Uninstalling networkx-3.6.1:
      Successfully uninstalled networkx-3.6.1
ERROR: pip's dependency resolver does not currently take into account all the packages that are installed. This behaviour is the source of the following dependency conflicts.
bigframes 2.42.0 requires tabulate>=0.9, but you have tabulate 0.8.10 which is incompatible.
mapclassify 2.10.0 requires networkx>=3.2, but you have networkx 2.8.8 which is incompatible.
spopt 0.7.0 requires networkx>=3.2, but you have networkx 2.8.8 which is incompatible.
scikit-image 0.25.2 requires networkx>=3.0, but you have networkx 2.8.8 which is incompatible.
momepy 0.11.0 requires networkx>=3.2, but you have networkx 2.8.8 which is incompatible.
Successfully installed Deprecated-1.2.18 networkx-2.8.8 tabulate-0.8.10 tsplib95-0.7.1 wrapt-1.17.3

```

Carga de librerías II

```
In [ ]: import urllib.request #Hacer llamadas http a paginas de la red
import tsplib95               #Modulo para las instancias del problema del TSP
import math                   #Modulo de funciones matematicas. Se usa para exp
import random                  #Para generar valores aleatorios
from google.colab import drive #Libreria para cargar archivos desde drive
drive.mount('/content/drive')
```

Mounted at /content/drive

Carga de los datos del problema

```
In [ ]: #http://elib.zib.de/pub/mp-testdata/tsp/tsplib/
#Documentacion :
# http://comopt.ifi.uni-heidelberg.de/software/TSPLIB95/tsp95.pdf
# https://tsplib95.readthedocs.io/en/stable/pages/usage.html
# https://tsplib95.readthedocs.io/en/v0.6.1/modules.html
# https://pypi.org/project/tsplib95/

#Descargamos el fichero de datos(Matriz de distancias)
#file = "swiss42.tsp" ;
#urllib.request.urlretrieve("http://comopt.ifi.uni-heidelberg.de/software/TSPLIB
#!gzip -d swiss42.tsp.gz      #Descomprimir el fichero de datos
file = '/content/drive/MyDrive/Maestria VIU/Algoritmos Optimizacion/Actividad gu
#print(contenido)

#Coordendas 51-city problem (Christofides/Eilon)
#file = "eil51.tsp" ; urllib.request.urlretrieve("http://comopt.ifi.uni-heidelbe

#Coordenadas - 48 capitals of the US (Padberg/Rinaldi)
#file = "att48.tsp" ; urllib.request.urlretrieve("http://comopt.ifi.uni-heidelbe
```

```
In [ ]: #Carga de datos y generación de objeto problem
#####
problem = tsplib95.load(file)

#Nodos
Nodos = list(problem.get_nodes())

#Aristas
Aristas = list(problem.get_edges())
```

```
In [ ]: print(problem)
```

NAME: swiss42
COMMENT: 42 Staedte Schweiz (Fricker)
TYPE: TSP
DIMENSION: 42
EDGE_WEIGHT_TYPE: EXPLICIT
EDGE_WEIGHT_FORMAT: FULL_MATRIX
EDGE_WEIGHT_SECTION:
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12 57 28 43 70 65 66 37 103 84 125 129 72 126 141 183 124
15 0 34 23 27 40 19 32 93 117 88 100 87 75 63 67 71 69 62 63 96 164 132 131 212 1
06 44 33 51 77 75 72 52 118 99 132 132 67 139 148 186 122
30 34 0 11 18 57 36 65 62 84 64 89 76 93 95 100 104 98 57 88 99 130 100 101 179 8
6 51 4 18 43 45 95 45 115 93 152 159 100 112 114 153 94
23 23 11 0 11 48 26 54 70 94 69 89 75 84 84 89 92 89 54 78 99 141 111 109 190 89
44 11 29 54 56 89 47 118 96 147 151 90 122 126 163 101
32 27 18 11 0 40 20 58 67 92 61 78 65 76 83 89 91 95 43 72 110 141 116 105 190 81
34 19 35 57 63 97 58 129 107 156 158 92 129 127 161 95
55 40 57 48 40 0 23 55 96 123 78 75 62 36 56 66 63 95 37 34 137 174 156 129 224 9
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37 32 65 54 58 55 45 0 124 149 118 126 113 80 42 42 49 40 87 60 94 195 158 163 24
2 135 65 63 79 106 101 50 66 118 104 109 103 36 160 178 218 153
92 93 62 70 67 96 85 124 0 28 29 68 63 122 148 155 156 159 67 129 148 78 80 39 12
9 46 82 65 55 40 61 157 97 159 135 212 221 159 110 72 95 35
114 117 84 94 92 123 111 149 28 0 54 91 88 150 174 181 182 181 95 157 159 50 65 2
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...]
```

```
In [ ]: #Probamos algunas funciones del objeto problem

#Distancia entre nodos
problem.get_weight(0, 1)

#Todas las funciones
#Documentación: https://tsplib95.readthedocs.io/en/v0.6.1/modules.html

#dir(problem)
```

Out[]: 15

Funcionas basicas

```
In [ ]: #Funcionas basicas
#####
```

```

#Se genera una solucion aleatoria con comienzo en en el nodo 0
def crear_solucion(Nodos):
    solucion = [Nodos[0]]
    for n in Nodos[1:]:
        solucion = solucion + [random.choice(list(set(Nodos) - set({Nodos[0]}) - set(
        return solucion

#Devuelve la distancia entre dos nodos
def distancia(a,b, problem):
    return problem.get_weight(a,b)

#Devuelve la distancia total de una trayectoria/solucion
def distancia_total(solucion, problem):
    distancia_total = 0
    for i in range(len(solucion)-1):
        distancia_total += distancia(solucion[i],solucion[i+1] , problem)
    return distancia_total + distancia(solucion[len(solucion)-1],solucion[0], problem)

sol_temporal = crear_solucion(Nodos)

distancia_total(sol_temporal, problem), sol_temporal

```

```
Out[ ]: (4721,
        [0,
          25,
          36,
          6,
          9,
          32,
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          20,
          41,
          4])
```

BUSQUEDA ALEATORIA

```
In [ ]: #####
# BUSQUEDA ALEATORIA
#####

def busqueda_aleatoria(problem, N):
    #N es el numero de iteraciones
    Nodos = list(problem.get_nodes())

    mejor_solucion = []
    #mejor_distancia = 10e100
    mejor_distancia = float('inf')

    #Inicializamos con un valor
    #Inicializamos con un valor
```



```

for i in range(N):
    solucion = crear_solucion(Nodos)
    distancia = distancia_total(solucion, problem)

    if distancia < mejor_distancia:
        mejor_solucion = solucion
        mejor_distancia = distancia

print("Mejor solución:" , mejor_solucion)
print("Distancia      :" , mejor_distancia)
return mejor_solucion

#Busqueda aleatoria con 5000 iteraciones
solucion = busqueda_aleatoria(problem, 10000)

```

Mejor solución: [0, 26, 12, 28, 18, 27, 20, 2, 33, 35, 36, 17, 15, 6, 41, 11, 31, 16, 34, 24, 13, 37, 32, 25, 30, 29, 1, 7, 14, 5, 10, 40, 39, 21, 8, 23, 3, 9, 38, 22, 19, 4]
 Distancia : 3653

BUSQUEDA LOCAL

```

In [ ]: #####
# BUSQUEDA LOCAL
#####
def genera_vecina(solucion):
    #Generador de soluciones vecinas: 2-opt (intercambiar 2 nodos) Si hay N nodos
    #Se puede modificar para aplicar otros generadores distintos que 2-opt
    #print(solucion)
    mejor_solucion = []
    mejor_distancia = 10e100
    for i in range(1,len(solucion)-1):
        for j in range(i+1, len(solucion)):

            #Se genera una nueva solución intercambiando los dos nodos i,j:
            # (usamos el operador + que para listas en python las concatena) : ej.: [
            vecina = solucion[:i] + [solucion[j]] + solucion[i+1:j] + [solucion[i]] +

            #Se evalua la nueva solución ...
            distancia_vecina = distancia_total(vecina, problem)

            #... para guardarla si mejora las anteriores
            if distancia_vecina <= mejor_distancia:
                mejor_distancia = distancia_vecina
                mejor_solucion = vecina
    return mejor_solucion

#solucion = [1, 47, 13, 41, 40, 19, 42, 44, 37, 5, 22, 28, 3, 2, 29, 21, 50, 34,
print("Distancia Solucion Inicial:" , distancia_total(solucion, problem))

nueva_solucion = genera_vecina(solucion)
print("Distancia Mejor Solucion Local:", distancia_total(nueva_solucion, problem))

```

Distancia Solucion Inicial: 3653
Distancia Mejor Solucion Local: 3386

```
In [ ]: #Busqueda Local:
# - Sobre el operador de vecindad 2-opt(funcion genera_vecina)
# - Sin criterio de parada, se para cuando no es posible mejorar.
def busqueda_local(problem):
    mejor_solucion = []

    #Generar una solucion inicial de referencia(aleatoria)
    solucion_referencia = crear_solucion(Nodos)
    mejor_distancia = distancia_total(solucion_referencia, problem)

    iteracion=0          #Un contador para saber las iteraciones que hacemos
    while(1):
        iteracion +=1      #Incrementamos el contador
        #print('#',iteracion)

        #Obtenemos La mejor vecina ...
        vecina = genera_vecina(solucion_referencia)

        #... y la evaluamos para ver si mejoramos respecto a lo encontrado hasta el
        distancia_vecina = distancia_total(vecina, problem)

        #Si no mejoramos hay que terminar. Hemos Llegado a un minimo Local(según nue
        if distancia_vecina < mejor_distancia:
            #mejor_solucion = copy.deepcopy(vecina)    #Con copia profunda. Las copias
            mejor_solucion = vecina                    #Guarda la mejor solución encont
            mejor_distancia = distancia_vecina

        else:
            print("En la iteracion ", iteracion, ", la mejor solución encontrada es:")
            print("Distancia      :" , mejor_distancia)
            return mejor_solucion

        solucion_referencia = vecina

sol = busqueda_local(problem )
```

En la iteracion 41 , la mejor solución encontrada es: [0, 5, 26, 10, 41, 23, 9, 40, 24, 21, 39, 22, 38, 34, 33, 20, 32, 31, 35, 36, 17, 37, 7, 1, 2, 3, 15, 16, 14, 19, 13, 11, 25, 12, 18, 6, 4, 8, 29, 30, 28, 27]
Distancia : 1649

SIMULATED ANNEALING

```
In [ ]: #####
# SIMULATED ANNEALING
#####

#Generador de 1 solucion vecina 2-opt 100% aleatoria (intercambiar 2 nodos)
#Mejorable eligiendo otra forma de elegir una vecina.
def genera_vecina_aleatorio(solucion):

    #Se eligen dos nodos aleatoriamente
    i,j = sorted(random.sample( range(1,len(solucion)) , 2))
```

```

#Devuelve una nueva solución pero intercambiando los dos nodos elegidos al azar
return solucion[:i] + [solucion[j]] + solucion[i+1:j] + [solucion[i]] + soluci

#Funcion de probabilidad para aceptar peores soluciones
def probabilidad(T,d):
    if random.random() < math.exp( -1*d / T) :
        return True
    else:
        return False

#Funcion de descenso de temperatura
def bajar_temperatura(T):
    return T*0.99

```

```

In [ ]: def recocido_simulado(problem, TEMPERATURA ):
    #problem = datos del problema
    #T = Temperatura

    solucion_referencia = crear_solucion(Nodos)
    distancia_referencia = distancia_total(solucion_referencia, problem)

    mejor_solucion = []          #x* del pseudocódigo
    mejor_distancia = 10e100     #F* del pseudocódigo

    N=0
    while TEMPERATURA > .0001:
        N+=1
        #Genera una solución vecina
        vecina =genera_vecina_aleatorio(solucion_referencia)

        #Calcula su valor(distancia)
        distancia_vecina = distancia_total(vecina, problem)

        #Si es la mejor solución de todas se guarda(siempre!!!)
        if distancia_vecina < mejor_distancia:
            mejor_solucion = vecina
            mejor_distancia = distancia_vecina

        #Si la nueva vecina es mejor se cambia
        #Si es peor se cambia según una probabilidad que depende de T y delta(distan
        if distancia_vecina < distancia_referencia or probabilidad(TEMPERATURA, abs(
            #solucion_referencia = copy.deepcopy(vecina)
            solucion_referencia = vecina
            distancia_referencia = distancia_vecina

        #Bajamos la temperatura
        TEMPERATURA = bajar_temperatura(TEMPERATURA)

    print("La mejor solución encontrada es " , end="")
    print(mejor_solucion)
    print("con una distancia total de " , end="")
    print(mejor_distancia)
    return mejor_solucion

sol = recocido_simulado(problem, 10000000)

```

La mejor solución encontrada es [0, 18, 12, 10, 8, 25, 11, 13, 15, 16, 19, 1, 3, 27, 28, 2, 4, 6, 26, 5, 14, 37, 7, 17, 31, 36, 35, 20, 33, 34, 32, 30, 29, 38, 2, 39, 24, 40, 21, 23, 41, 9]
con una distancia total de 1707

In []: